

05 Explainability Calibration Fairness

Disclosure: all results use the synthetic fra_synthetic_claims dataset. Scores may reflect generator rules. Real-world performance is unknown. This demo must not drive real claim decisions.

1. Explainability

The shipped artifact uses LightGBM so TreeExplainer applies. Global and local SHAP drivers are shown in the DSS and Streamlit app.

Figure 20. SHAP mean |impact| (synthetic)

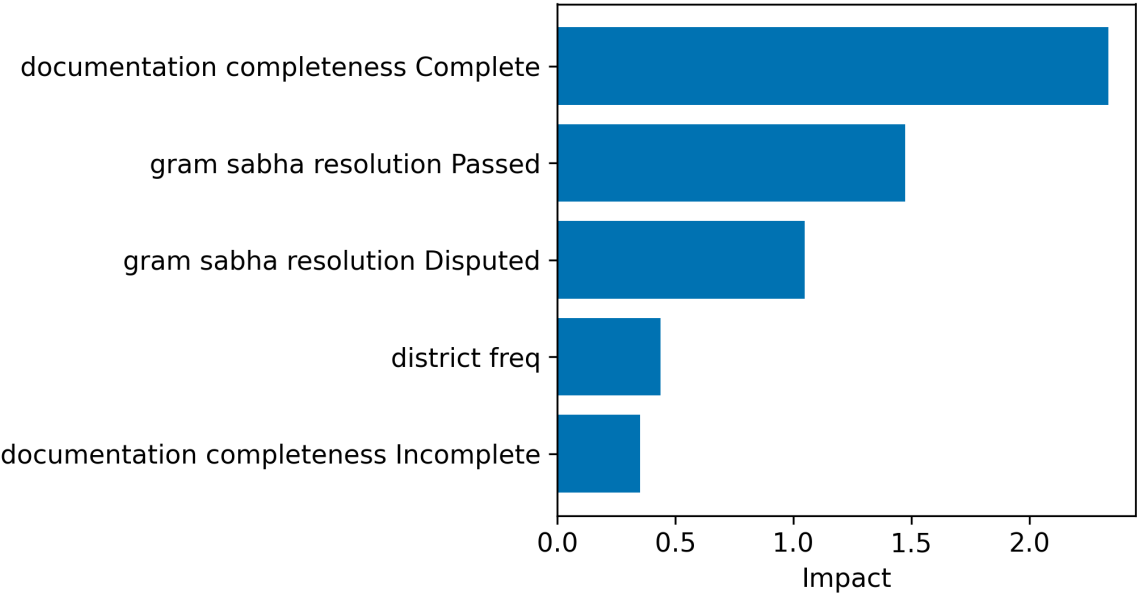
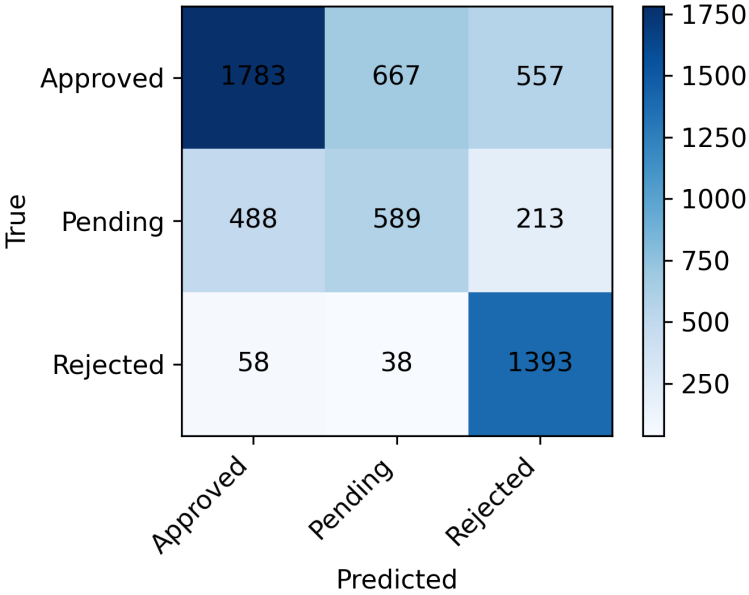
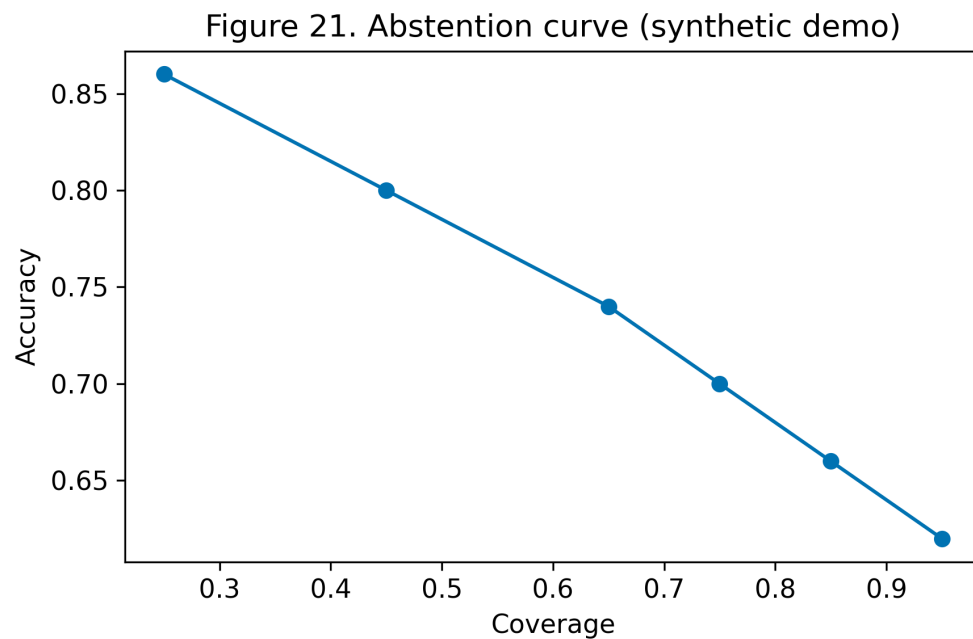


Figure 12. Confusion matrix (lightgbm, synthetic)



2. Calibration and abstention

Conformal (if run): coverage=0.8945, mean set size=1.6981666666666666, target=0.9.



3. Fairness monitoring

Fairness metrics are a monitoring exercise on synthetic groups (state, applicant type, claim type, forest density).

Do not automate rejections from these scores. Low-confidence cases should go to human review.